

Real-time data collection and monitoring: Continuous tracking of vitals (heart rate, glucose, oxygen saturation, etc) enables early detection of anomalies and proactive interventions.

Remote accessibility: Patients and clinicians should be able to access health data anytime, anywhere. Remote accessibility supports telemedicine, home care, and chronic disease management.

Scalability and flexibility: Systems can expand from a few devices to thousands across hospitals or patient populations. A modular design allows the integration of new sensors, apps, or analytics engines.

Data analytics and intelligence: AI/ML models should be applied to IoT data for predictive insights (e.g., risk of cardiac arrest). Data analytics enables personalised treatment plans based on continuous data streams.

Key applications of IoT in healthcare industry

The integration of IoT devices into the healthcare industry is revolutionising patient care, improving medical outcomes, and streamlining operations.

Remote patient monitoring: IoT-based remote patient monitoring uses smartwatches, smart inhalers, connected pacemakers, blood glucose monitors, phones and other monitoring devices to collect patient health data automatically. These devices allow for continuous monitoring of vital signs, temperature, blood pressure, body fat percentage, etc, to enable real-time data collection and analysis.

For example, patients with chronic conditions like diabetes or hypertension receive constant monitoring from the comfort of their homes, reducing the need for frequent doctor visits and hospitalisations. This not only enhances the quality of care but also facilitates a shift to more cost-effective and accessible healthcare, especially for underserved and remote populations.

Mobile health: IoT devices can

send patients' medical or physical data to the smartphones of healthcare providers using a healthcare app. Doctors can swiftly schedule online appointments with this application.

Electronic health records: Doctors can acquire real-time patient reports using IoT-based health systems. IoT in healthcare allows patients to contact their doctors at any time.

Real time health systems (RTHS): RTHS simplifies operational, clinical, and administrative teamwork, improving healthcare services.

IoT heart rate monitors track a person's heart rate continually. IoT-based health monitors can measure Parkinson's disease symptoms and their severity throughout the day. Digestible sensors help to measure the stomach's pH and check for any internal bleeding.

Emergency care: IoT devices alert rescue workers to provide faster and better medical care by collecting or sending data in real time.

Hospital automation: IoT streamlines hospital workflows through smart beds, asset tracking, and automated inventory management. It enhances safety, reduces costs, and improves staff efficiency in hospitals.

Future of IoT in healthcare

AIoT (AI + IoT): AI will be used to analyse IoT data in real-time, enabling predictive diagnostics and personalised

therapies. This will lead to faster, more accurate diagnoses from real-time data streams (e.g., predicting asthma attacks or cardiac events).

Digital twins of patients: Virtual replicas of individuals will help in predictive modelling of hospital operations to optimise staffing, bed allocation, and emergency response.

5G-enabled healthcare: Ultra-low latency networks will power real-time telemedicine, robotic surgeries, and AR/VR-assisted care. This will provide affordable healthcare for rural and underserved areas where connectivity is currently a barrier.

Voice-assisted smart devices: This will enable hands-free interaction in surgical rooms or senior care.

Blockchain for health data security: Decentralised health record management will enable better trust in them.

IoT in mental health: IoT systems will help monitor sleep, behaviour, and speech patterns to support mental wellness.

IoT in healthcare is reshaping the healthcare delivery model. Its data-driven nature provides actionable insights and supports caregivers with timely information, among other things. IoT-based solutions enable a proactive approach where continuous monitoring, predictive analytics, and real-time alerts help anticipate health issues instead of waiting to respond to acute episodes. **END** 

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